# Transfer task: plan and begin



**antocciphysics | TEACHER KEY | CPA**

**Learning target:** Organize the given evidence and begin an independent position prediction.

**Suggested first pass:** 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

**01. Bring back an earlier idea (retrieve)** *Review: Unit 1, Day 18; 3 course days later.*
12 N left; 15-3-12=0 N.

**02. Explain a worked example (worked)**
A 10 m rightward path; later timed position observations can test equal motion per equal time.

**03. Organize the givens (model)**
Correct table and sketch; initial x negative, velocity components positive; time in seconds, velocity km/s.

**04. Find the central end point (practice)**
Δx 131, 360, 000 km; xf 11, 360, 000 km;Δy=yf 35, 200, 000 km.

**05. Start the bounds (reason)**
x lower 8, 270, 000 km; upper 14, 450, 000 km; y lower 34, 400, 000 km; upper 36, 000, 000 km. Bounds are givens for this CPA model, not derived trig bounds.

**06. Choose a method (mixed)** *Review: Unit 1, Day 9; 12 course days later.*
The straight-moving rover can have zero net force. The turning rover changes velocity direction and needs net force.

**07. Checkpoint (exit)**
Assumption constant velocity; gravity/Earth motion omitted. Require visible independent work and specific next step.

**08. STRETCH: Apply it another way (stretch)**
Example: initial speed was lower, or object slowed; multiple timed positions distinguish initial velocity and acceleration.

**Teaching note:** Independent transfer task, Day 1 of 2.CPA uses provided components and component bounds; Honors resolves a stated angle and propagates speed bounds. Use each track's own givens.

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**01. Bring back an earlier idea (retrieve)** *Review: Unit 1, Day 18; 3 course days later.*
At rest T=29.4 N. Accelerating: T=m(g+a)=35.4 N.

**02. Explain a worked example (worked)**
A 10 m rightward path; later timed position observations can test equal motion per equal time.

**03. Plan and represent (model)**
Axes fixed,+x initially Earthward,+y perpendicular; asteroid negative x. Plan must start consistent units.

**04. Central prediction (practice)**
v=(8.2103695, 2.1999619) km/s; xf 11, 365, 912.4 km; yf 35, 199, 390.1 km. Displacement=(131, 365, 912.4, 35, 199, 390.1) km.

**05. Propagate bounded inputs (reason)**
δx=50, 000+(.2 cos 15)(16, 000, 000)=3, 140, 962.6 km;δy=(.2 sin 15)(16, 000, 000)=828, 220.9 km. x 8, 224, 949.7 to 14, 506, 875.0; y 34, 371, 169.2 to 36, 027, 611.1 km.

**06. Choose a method (mixed)** *Review: Unit 1, Day 9; 12 course days later.*
Tangent velocity direction changes, so acceleration and net force are inward and nonzero. Constant speed is insufficient for equilibrium.

**07. Checkpoint (exit)**
At 0:-120, 000, 000 central, range[-120, 050, 000,-119, 950, 000]. At final time use 2-3. Constant velocity; gravity/Earth motion omitted.

**08. STRETCH: Apply it another way (stretch)**
v=1.6 m/s; or a=2×8/25=0.64 m/s² from rest. Intermediate positions distinguish the models; original 10 m is outside the stated bound.

**Teaching note:** Independent transfer task, Day 1 of 2.CPA uses provided components and component bounds; Honors resolves a stated angle and propagates speed bounds. Use each track's own givens.